CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 4: Quadratic Equations

Previous year board questions, topic-wise: 22 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Word problems

1. Over a 200 km journey, a fast train takes one hour less than a slow train. The slow train runs 10 km/h slower than the fast one. Find the speed of each train. CBSE 2026, 5 marks

2. Two squares have a total area of 640 m², and their perimeters differ by 64 m. Find the side of each square. CBSE 2026, 5 marks

3. A right-angled triangle has a perimeter of 60 cm and a hypotenuse of 25 cm. Find the lengths of the other two sides. CBSE 2025, 5 marks

4. A train covers 480 km at a uniform speed. Had it travelled 8 km/h slower, the journey would have taken 3 hours longer. Find the speed of the train. CBSE 2025, 5 marks

5. A rectangular floor is covered exactly by 200 square tiles. With tiles whose side is 1 unit longer, only 128 tiles are needed. Let the side of an original tile be x units. (i) Form a quadratic equation in x. (ii) Write it in standard form. (iii) Find x by factorisation, OR solve the equation by the quadratic formula. CBSE 2024, 4 marks, case study

6. A school photo is 18 cm long and 12 cm wide. Its length and width are each to be increased by x cm so that the area doubles. (i) Write an equation for this. (ii) Write it as a quadratic equation in standard form. (iii) Find the new length and width, OR decide whether any rational value of x gives a new area of 220 cm². CBSE 2023, 4 marks

7. Rehan's age 5 years ago multiplied by his age 7 years from now is one more than twice his present age. Find his present age. CBSE 2022 Term 2, 2 marks

8. Two taps together fill a tank in $1\frac{7}{8}$ hours. Working alone, the larger tap takes 2 hours less than the smaller one. How long does each tap take on its own? CBSE 2019, 4 marks

9. A plane took off 30 minutes late. To reach its destination 1500 km away on time, it flew 100 km/h faster than usual. Find its usual speed. CBSE 2018, 3 marks

10. A motor boat has a speed of 18 km/h in still water. It takes 1 hour longer to go 24 km upstream than to come back the same distance downstream. Find the speed of the stream. CBSE 2018, 4 marks

11. A train covers 63 km at a certain average speed and then 72 km at a speed 6 km/h faster. The whole journey takes 3 hours. Find the first speed. CBSE 2018, 4 marks

12. A motor boat has a speed of 24 km/h in still water. It takes 1 hour longer to go 32 km upstream than to return the same distance downstream. Find the speed of the stream. CBSE 2016, 4 marks

2 Nature of roots

13. The equation $ax^2 + bx + c = 0$, $a \neq 0$, has real and equal roots. Which relation holds? (a) $a = \frac{b^2}{c}$ (b) $b^2 = ac$ (c) $ac = \frac{b^2}{4}$ (d) $c = \frac{b^2}{a}$ CBSE 2024, 1 mark

14. What is the least positive value of k for which $2x^2 + kx - 4 = 0$ has rational roots? (a) $\pm 2\sqrt{2}$ (b) 2 (c) ± 2 (d) $\sqrt{2}$ CBSE 2023, 1 mark

15. For what value of m does $(m-1)x^2 + 2(m-1)x + 1 = 0$ have two equal real roots? CBSE 2022 Term 2, 2 marks

16. For what values of k does $x^2 + 4x + k = 0$ have real roots? CBSE 2019, 1 mark

17. The roots of $3x^2 - 10x + k = 0$ are reciprocals of each other. Find k. CBSE 2019, 1 mark

18. -5 is a root of $2x^2 + px - 15 = 0$, and the equation $p(x^2 + x) + k = 0$ has equal roots. Find k. CBSE 2016, 2 marks

3 Solving by factorisation

19. Solve for x: $\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$. CBSE 2022 Term 2, 2 marks

20. $x = 3$ is a root of $x^2 - 2kx - 6 = 0$. Find k. CBSE 2018, 1 mark

21. Solve for x: $\frac{1}{(x-1)(x-2)} + \frac{1}{(x-2)(x-3)} = \frac{2}{3}$, where $x \neq 1, 2, 3$. CBSE 2016, 3 marks

4 Solving by the quadratic formula

22. Solve for x: $\frac{1}{x+1} + \frac{2}{x+2} = \frac{4}{x+4}$, where $x \neq -1, -2, -4$. CBSE 2016, 4 marks

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